Data-Driven Policymaking in Smart Cities:
A Framework for Large-Scale Active Mobility Experiments

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Urban Mobility Context

- 20th Century's priority: motorized individual transport

- 21st Century: urge to foster healthier cities
  - public transport +
  - active mobility (walking and cycling)

Bicycle use around the world

Cycling has many advantages towards healthier cities

But use varies largely:

- Amsterdam: 38%

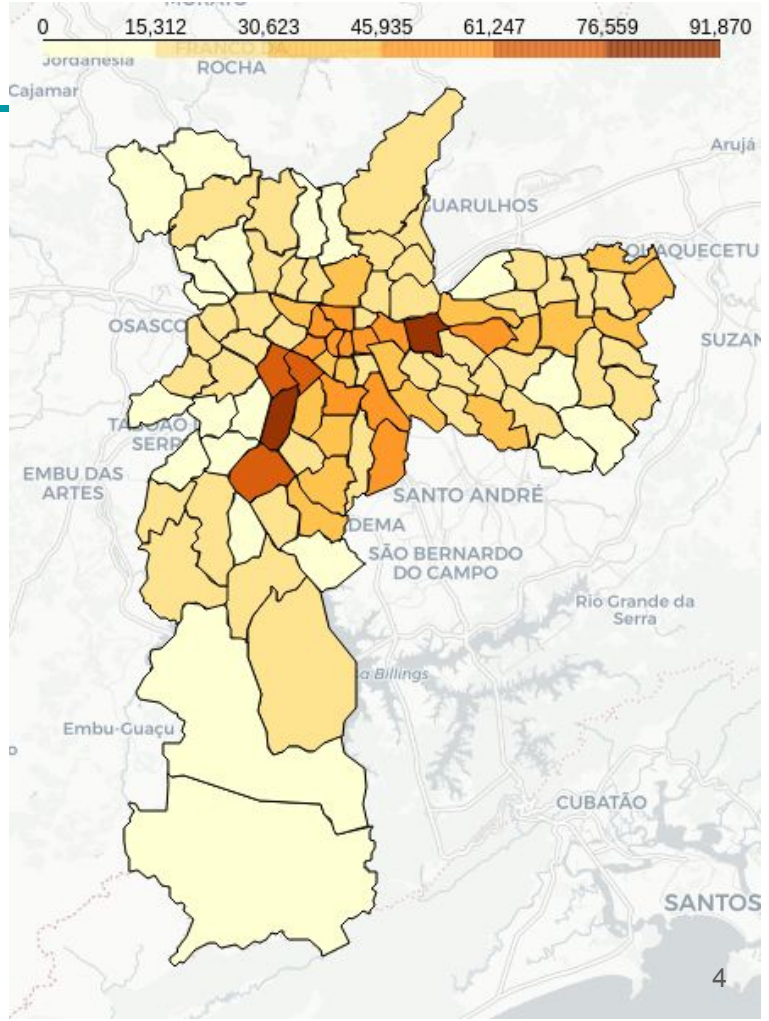
- Brazil
  - Santos: 15%
  - Rio de Janeiro, Porto Alegre: 5%

São Paulo Metropolitan Area

- Largest metropolitan area in the Western Hemisphere
- 22M people /17M motorized vehicles

- Low use of cycling
  - Daily commutes:
    2007: 0,6%
    2017: 0,8%
    2023: 1,1%

- How to speed up adoption???



Barriers to bicycle adoption

- **Public safety**
- **Road safety (sharing lanes w/ cars)**
- **Topography and weather (less relevant)**

Considerable investments in infrastructure:

2011: 60 km
2016: 468 km
2026: 757 km

But cycling growth is still very slow…





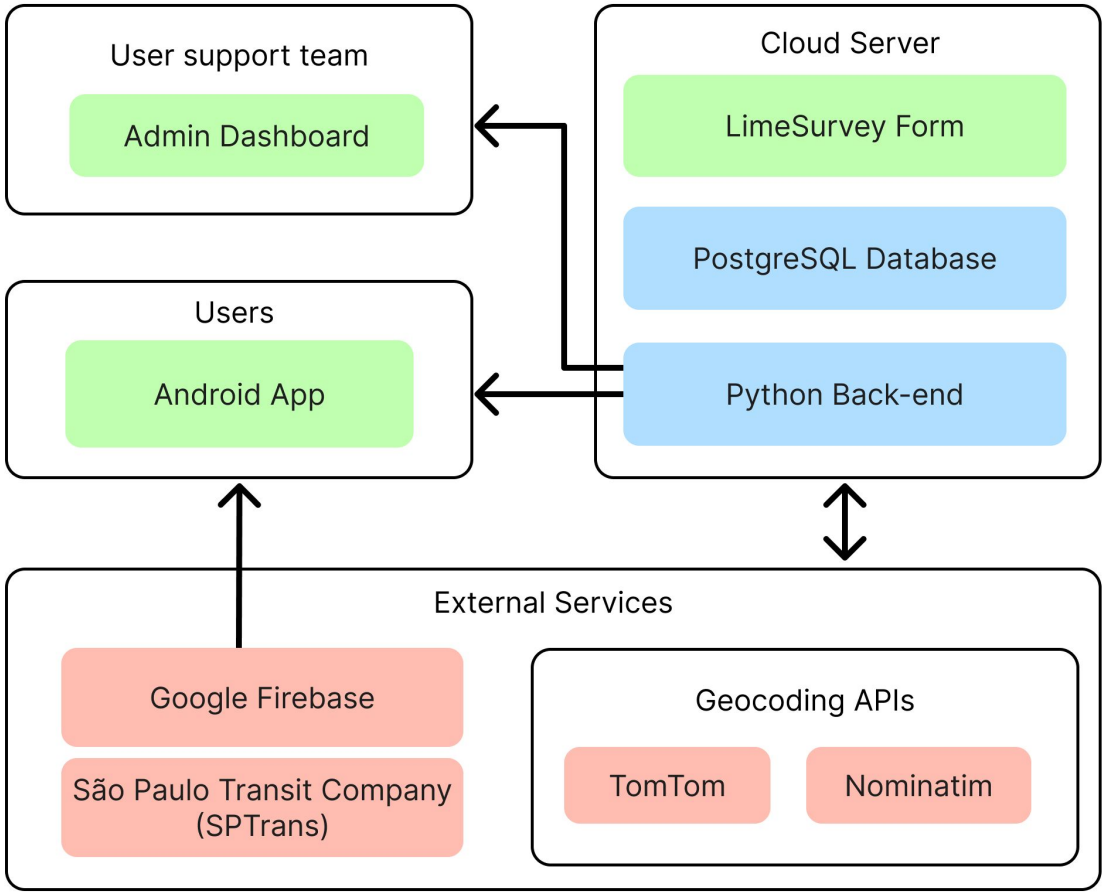
Financial Incentives

- Some initiatives around the world, mostly in Europe, tried different models.

- São Paulo approved a law determining a payment per Km for commuting to work, but never implemented it.

- City government asked our research group to design the implementation of this public policy based on scientific evidence.

Framework Architecture



Security

- authentication
- access control
- encryption
- app validation

User flow

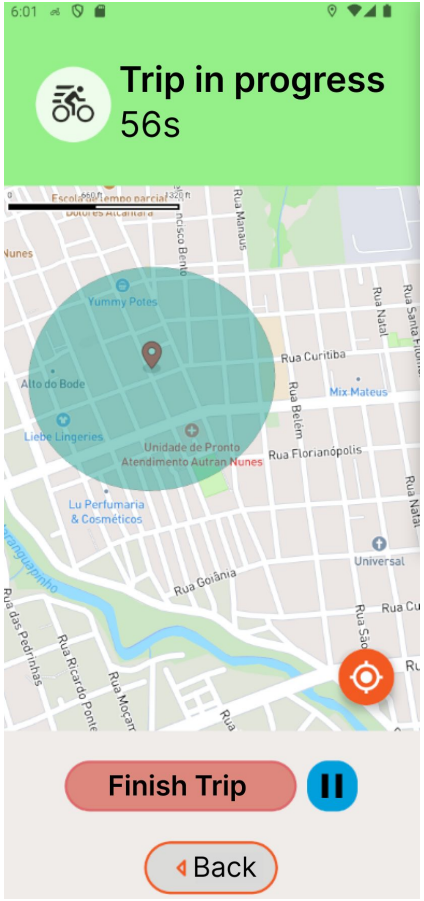
1. Fill in registration form
   - socio-demographic info
   - trip destinations (e.g., home, work, and school)

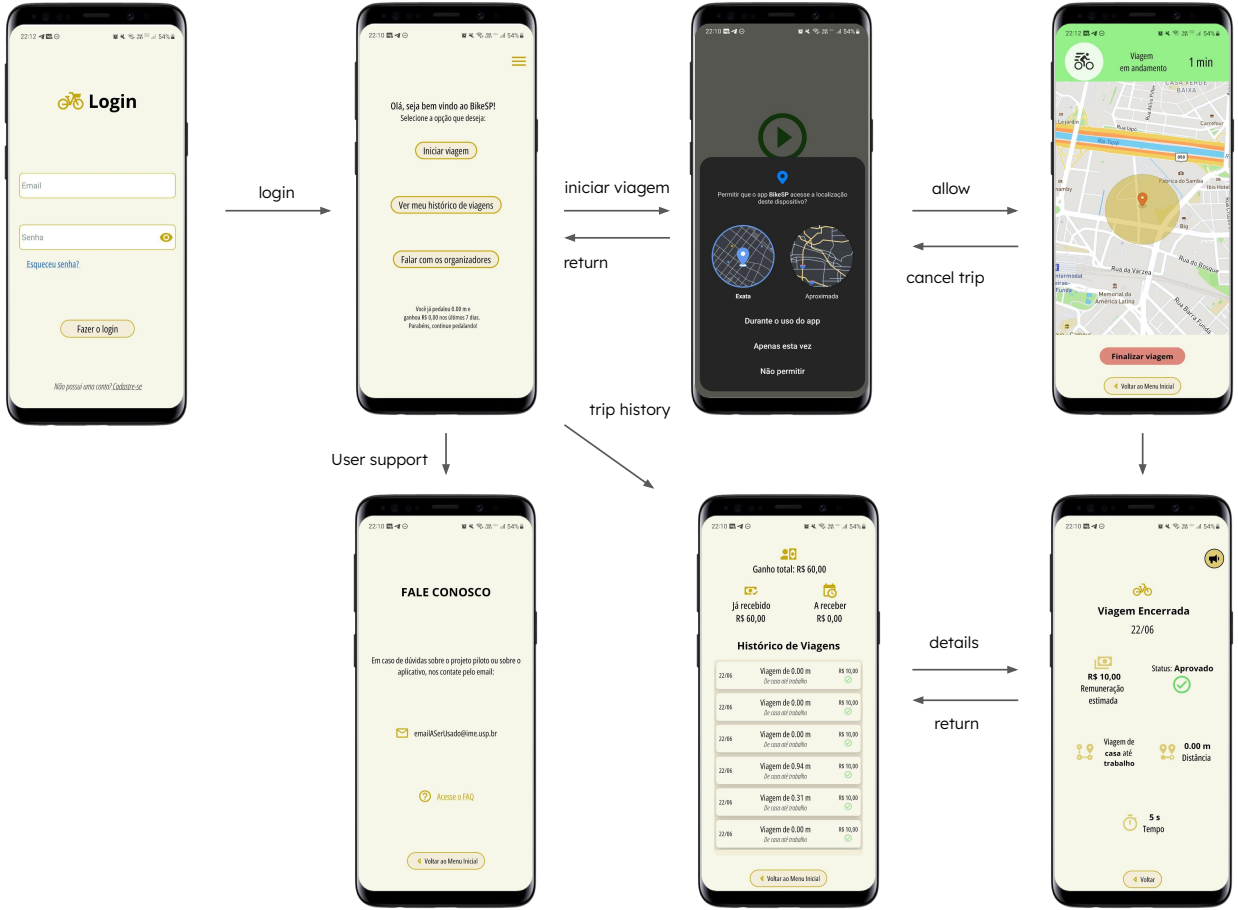
2. Tracks trips between destinations
   - smart heuristics to detect cycling

3. Compute payment based cycled kms
   - experimental cohorts receive different $$$
   - changed weekly to assess impact

4. Get weekly credit deposits in the public transit card

The Mobile Application

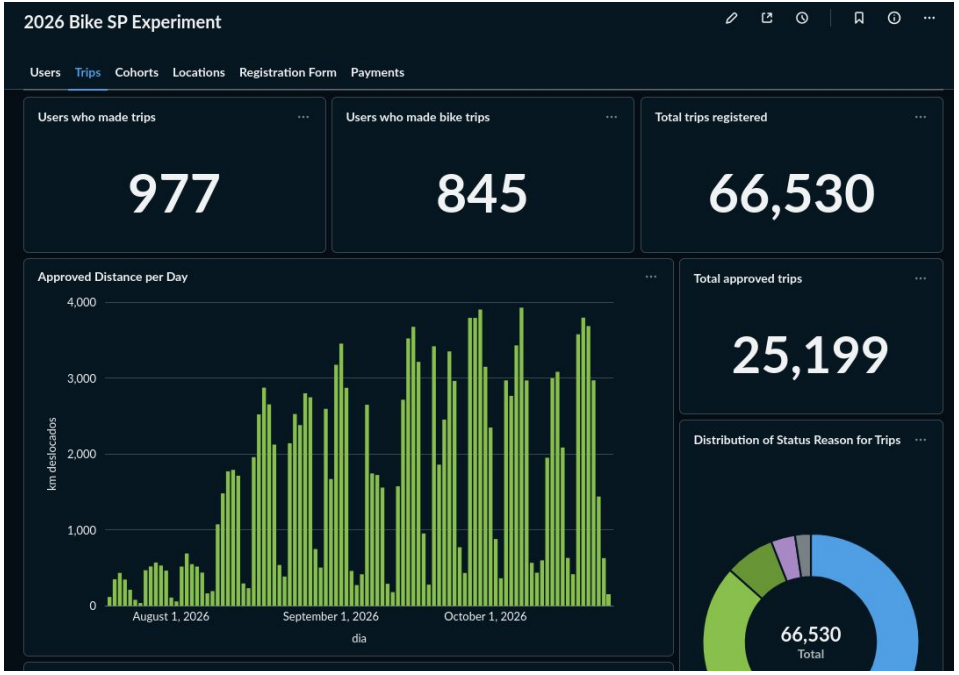


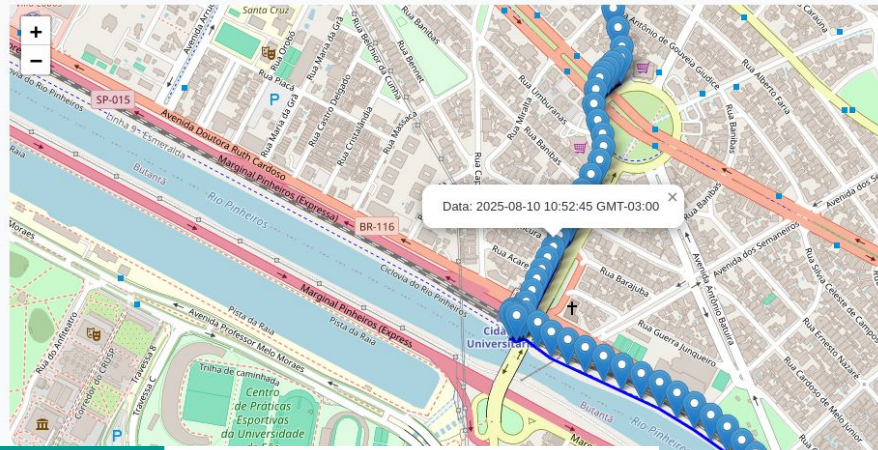


Operational Dashboards

Admin dashboard

Live Monitoring Dashboard







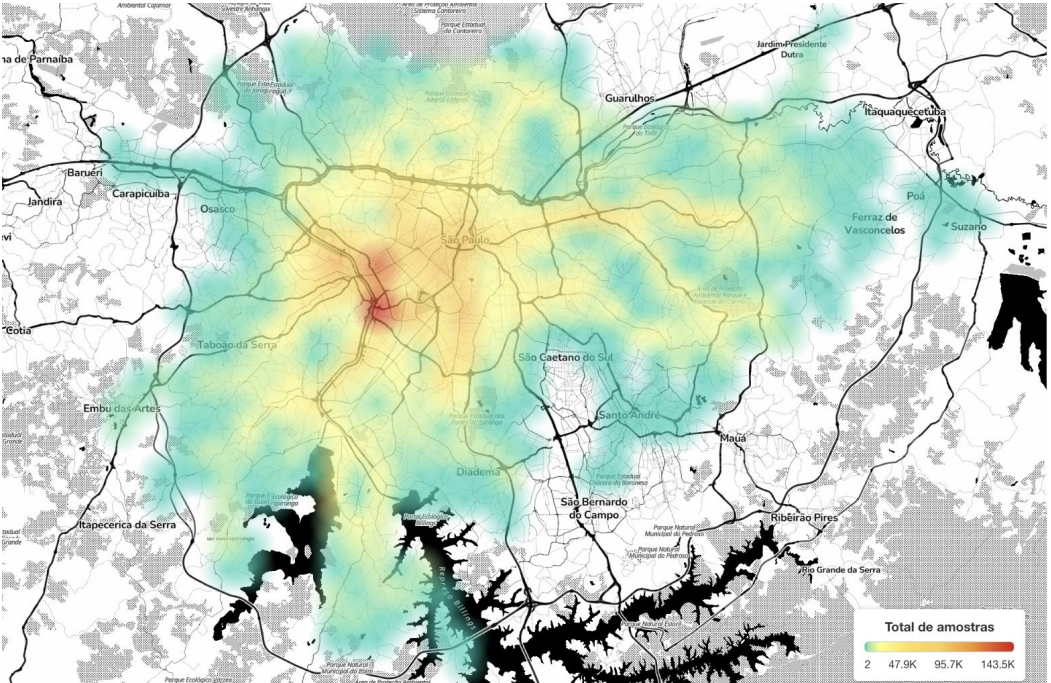
2025 Experiment in São Paulo

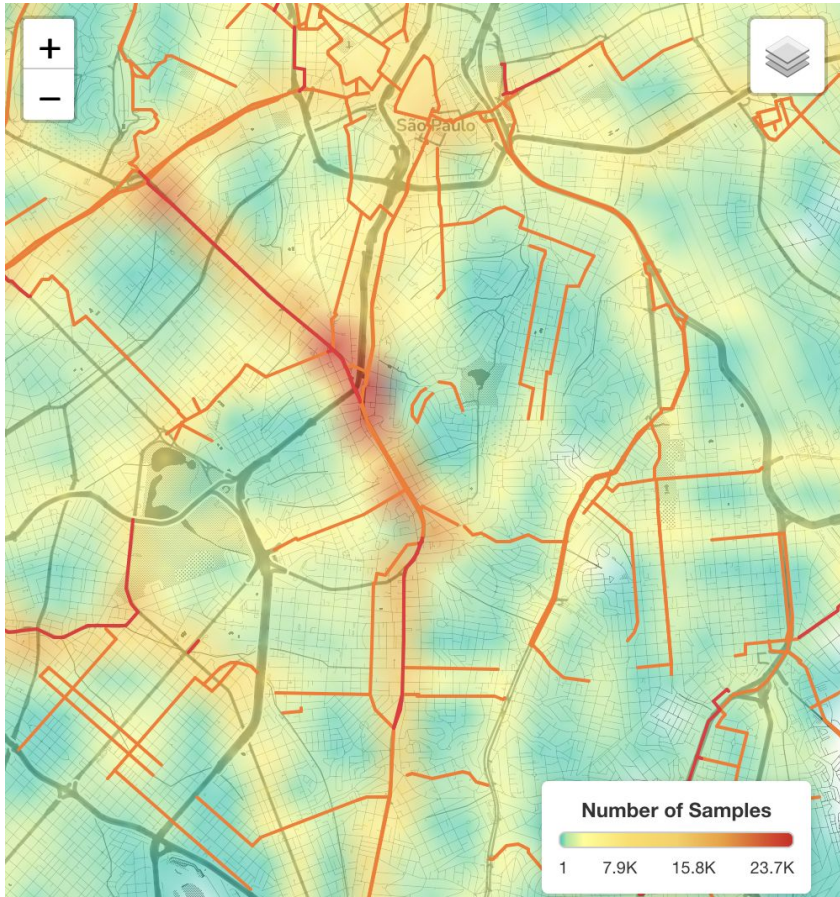
- data collection over 8 weeks
- 1217 participants
- at most 2 compensated bike trips per weekday
- each week, randomly selected each user's cohort:
  - R$ 0,00/km, R$ 0,32/km, R$ 0,63/km

- collected 30K+ cycling trips
- Validated the software and the operational process
- Results about the real impact on cycling were inconclusive (registration bias)

Analytics Dashboard

bikescienceweb.interscity.org





Experiments in Fortaleza

- Tailored, smaller-scale version of the experiment
- investigate effect of diversified incentive structures (e.g. direct cash transfers and cinema tickets)
- Connection to public bike-sharing stations

- Improvements to the software framework:
  - configuration-driven build system
  - dynamic reconfiguration based on variables stored in DB

Ongoing 2026 São Paulo Experiment

- On the 2025 experiment, tracking trips was manual.

- We redesigned the experiment so the app tracks trips as a background task to reduce registration bias.

- Now running with **1,090 users** for three months.
- Up to now, we recorded:
  - **68,927 total trips**
  - **26,078 validated bike trips**

Policy Brief

Data Analysis:

- Impact of different levels of financial incentive on the adoption of cycling
- Impact on non-cycling trips (car and public transport)

Objective: write a policy brief advising the city government on how to best regulate the law on cycling credits.

- Target demographics
- Compensation limits
- Amount paid per km

Conclusions

- A generic framework for active mobility experiments is valuable for policy-making
- Data quality is essential for informing public policy with rigorous scientific evidence
- Our extensible, open-source solution is available for experiments in other cities by other research groups, we will be happy to collaborate :-)

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gitlab.com/interscity/bikesp

interscity.org/bikesp



